

Numerai Signals:

Development Status

Focusing on Lessons Learned & Personal Experience

Why Numerai Signals?

Reasons to Participate in Numerai Signals

- High Flexibility & Broad Skill Acquisition:
 - Allows usage of custom-built, scraped, or purchased datasets.
 - Covers the entire pipeline: from data ingestion and feature engineering to modeling.
 - **Higher Potential Payouts** compared to the standard Tournament.
- Ideal for individuals unable to invest personal capital due to:
 - Strict compliance/company regulations on personal stock trading.
 - Lack of capital required to execute large-scale quantitative strategies.
 - Skyrocketing prices of quality stocks making them hard to purchase individually (e.g., Kioxia, Furukawa Electric, etc.).

What is Numerai Signals?

How It Differs from Traditional Competitions

- **Mechanism:** Users submit proprietary stock prediction signals, and are evaluated on how much value they add to Numerai's live fund operations.
 - **Data Privacy:** Users can utilize their own datasets without needing to upload the raw data itself (only the resulting signals are submitted).
 - **Low Barrier to Entry:** Submissions can be made without staking, allowing users to benchmark their signal strength risk-free.
- **Value of Orthogonality:** Value is derived from providing information that Numerai does not already possess, rather than just achieving a high correlation with raw returns.
- **Evaluation Metrics:** Evaluated using Numerai's unique metrics: Alpha and MPC (detailed on the next slide).
 - Note: Signals with high correlation to raw returns can sometimes yield a negative Alpha.
 - Success requires unique insights rather than just predicting price direction based on common market data.

Evaluation Metrics

Key Metrics Driving Payouts

- Alpha
 - Measures whether a user's signal contributes to return predictions in a way that Numerai can realistically execute.
 - High-performing signals that rely solely on small-cap or illiquid stocks are impractical for a large-scale fund to trade.
 - Uses a target metric called “target_chili” (Calculation details omitted; explained in the official blog: <https://blog.numer.ai/signals-alpha-and-mpc/>)
- MPC
 - Measures the marginal improvement in the overall portfolio performance when the user's signal is integrated.
 - Scores are strictly "blind" and unknown until the submission is processed.
- Summary
 - Alpha reflects pure signal performance, while MPC reflects portfolio-level contribution (two distinct properties).

My Approach

From Data Acquisition to Model Generation

- Utilized Data:
 - Numerai provided base data.
 - J-Quants API** (For comprehensive Japanese equity data).
- Modeling Pipeline:
 - Feature Transformation:** Ranked features to normalize distributions.
 - Model Selection:** Preferred non-linear models like **LightGBM** over traditional linear models to capture complex market interactions.

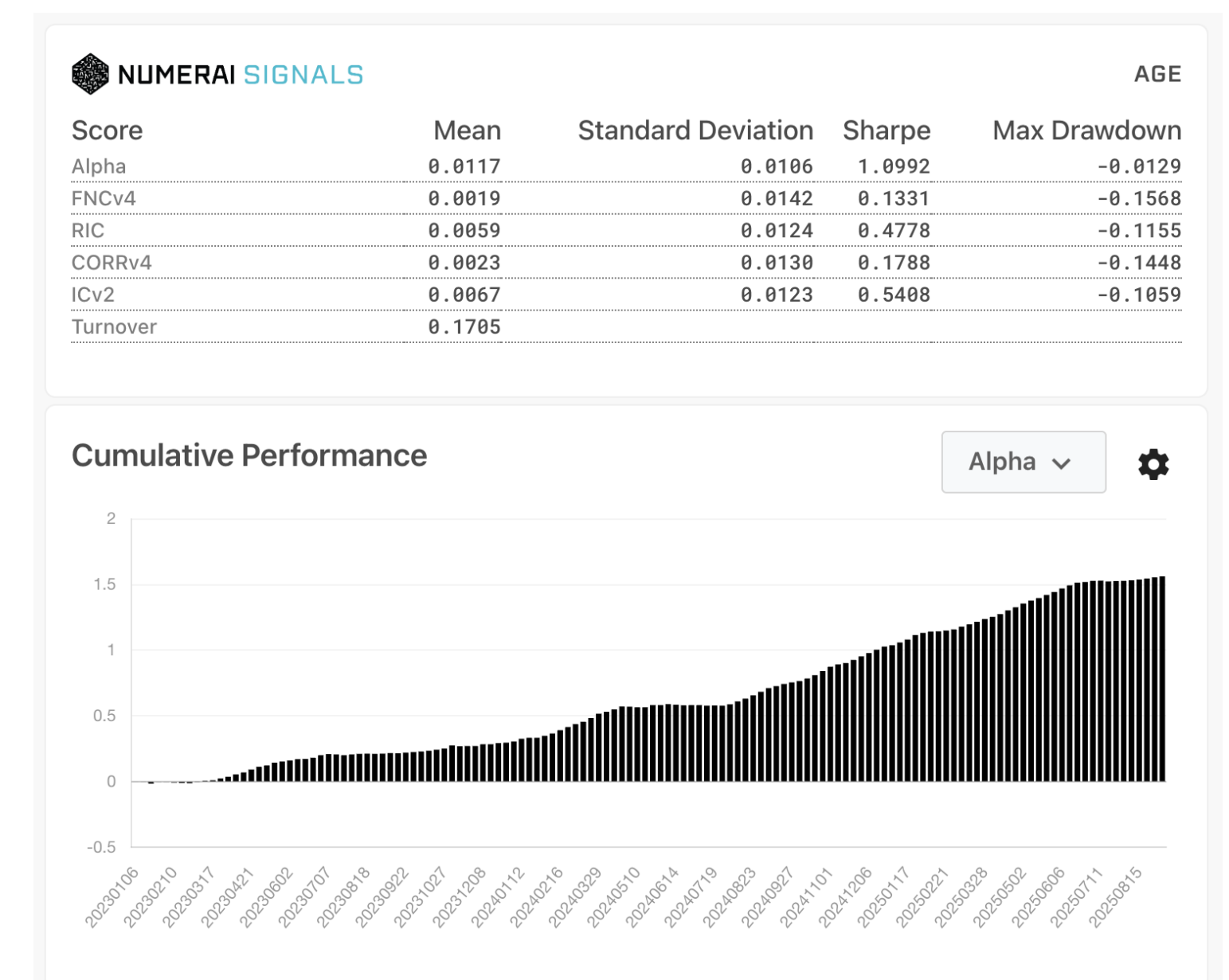
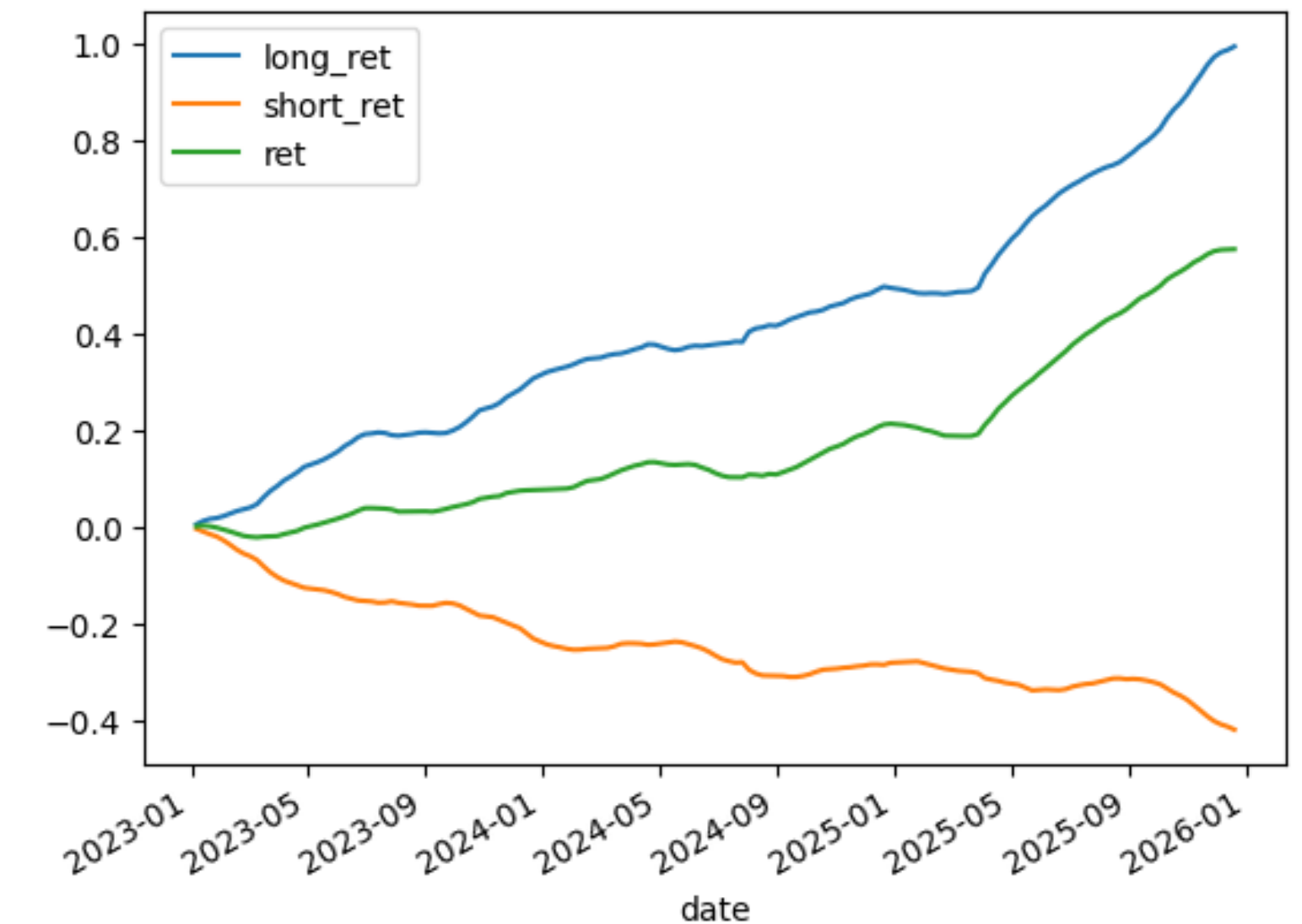
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7089743	ZYME US	US	BBG019XSYC89	2026-05-28	live	0.371494	0.475067

	Date	Code	Price	Vo_real	O	H	L	C	Vo
0	2008-05-07	13010	176.0	302000.0	1730.0	1770.0	1730.0	1760.0	30200.0
1	2008-05-08	13010	180.0	403000.0	1740.0	1800.0	1740.0	1800.0	40300.0
2	2008-05-09	13010	176.0	321000.0	1800.0	1800.0	1760.0	1760.0	32100.0
3	2008-05-12	13010	177.0	178000.0	1760.0	1770.0	1750.0	1770.0	17800.0
4	2008-05-13	13010	175.0	407000.0	1780.0	1780.0	1710.0	1750.0	40700.0
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15808054	2026-05-29	99870	5285.0	194500.0	5216.0	5335.0	5216.0	5285.0	194500.0
15808055	2026-05-29	99890	3627.0	841000.0	3679.0	3692.0	3627.0	3627.0	841000.0
15808056	2026-05-29	99900	720.0	95400.0	716.0	724.0	714.0	720.0	95400.0
15808057	2026-05-29	99910	1511.0	22400.0	1529.0	1532.0	1511.0	1511.0	22400.0
15808061	2026-05-29	99970	925.0	1800400.0	916.0	925.0	915.0	925.0	1800400.0

Initial Model

Submitting Raw Scores

- **Universe:** Restricted strictly to the Japanese equity universe.
- **Strategy:** Blended raw features (highly correlated with historical returns) with model predictions.
- **Results:** Performance was mediocre.
 - Alphas were higher when using Numerai's global data across the full universe.
- **Rationale for Deployment:** Deployed under the assumption that a Japan-specific model would offer a safer exit strategy during major market downturns.

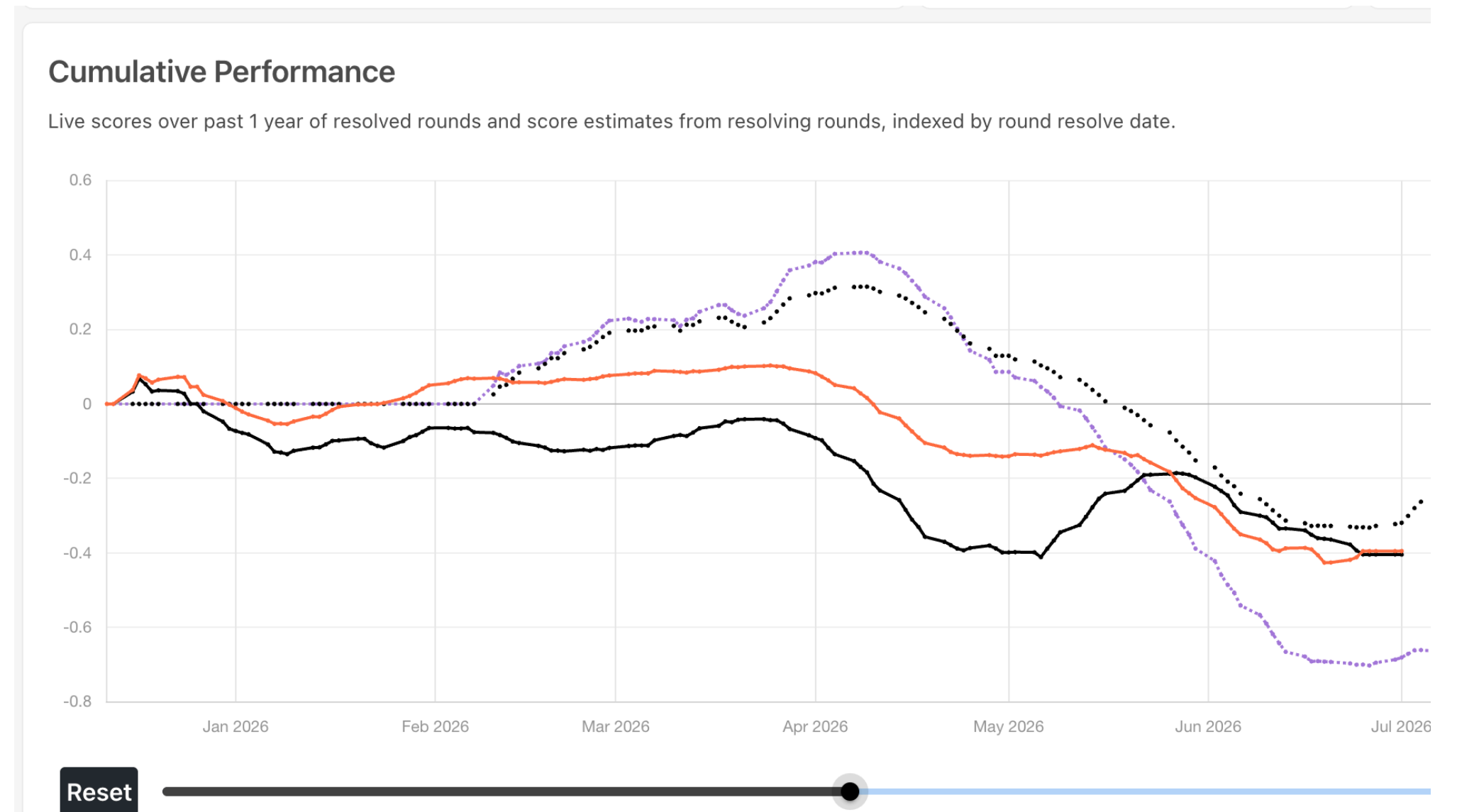


Initial Model Results

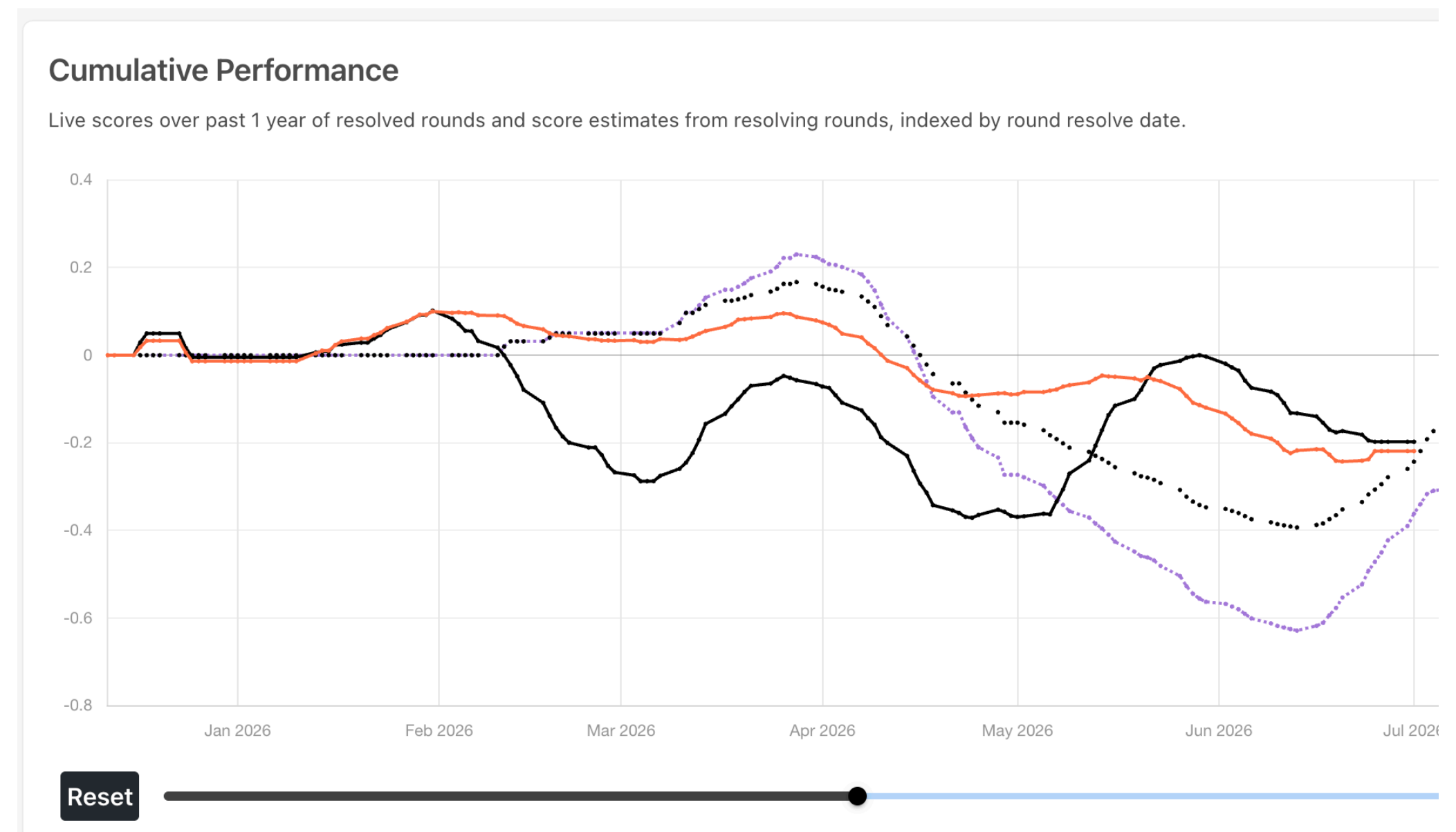
Performance Post-Deployment

- **Raw Features + Predictions:**
 - Underperformed significantly immediately after live deployment.
- **Raw Features Only:** Suffered severe initial losses, though it has shown a recent recovery.
 - The recent bounce aligns closely with the overall recovery of the Japanese stock market.
- **Key Flaw:** The model carried an unsustainably high exposure to the Japanese market factor.

- **Raw Features + Predictions:**



- **Raw Features Only**

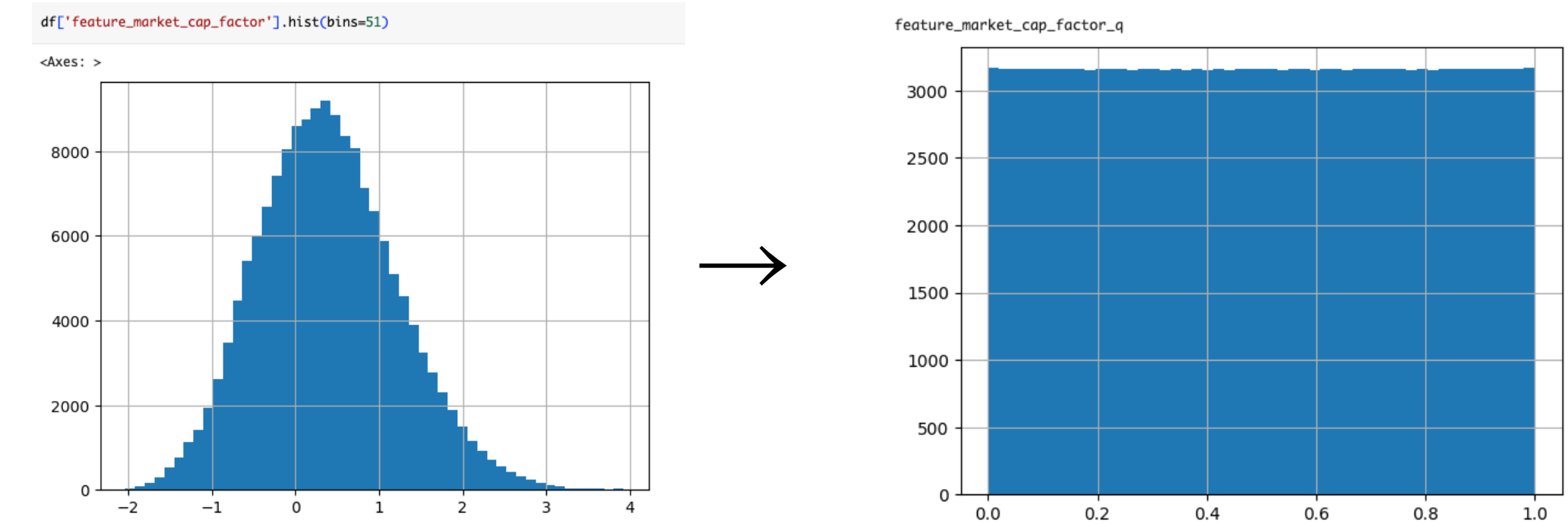


Current Model

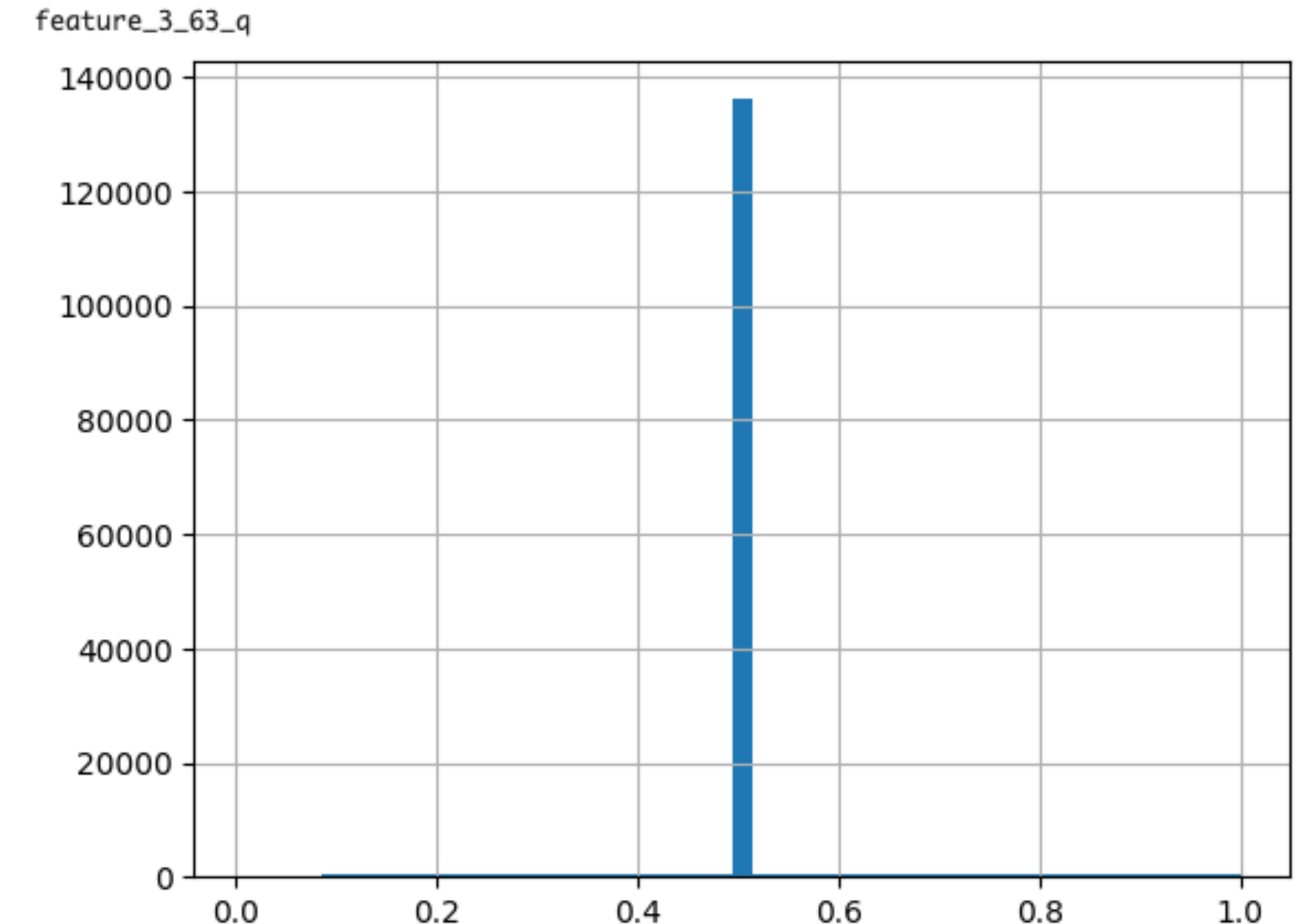
Model Improvements & Iterations

- **Data Integration:** Incorporated data provided directly by Numerai.
- **Feature Engineering (Ranking):** Transformed raw skewed features into uniform distributions bounded between 0 and 1.
- Handling Missing Global Data:
 - **Challenge:** Feature data was only available for the Japanese equity universe.
 - **Solution:** Features were ranked from 0.0 to 1.0. For all tickers outside the Japanese universe, missing values were imputed with a neutral baseline of 0.5.

特徴量のランク化



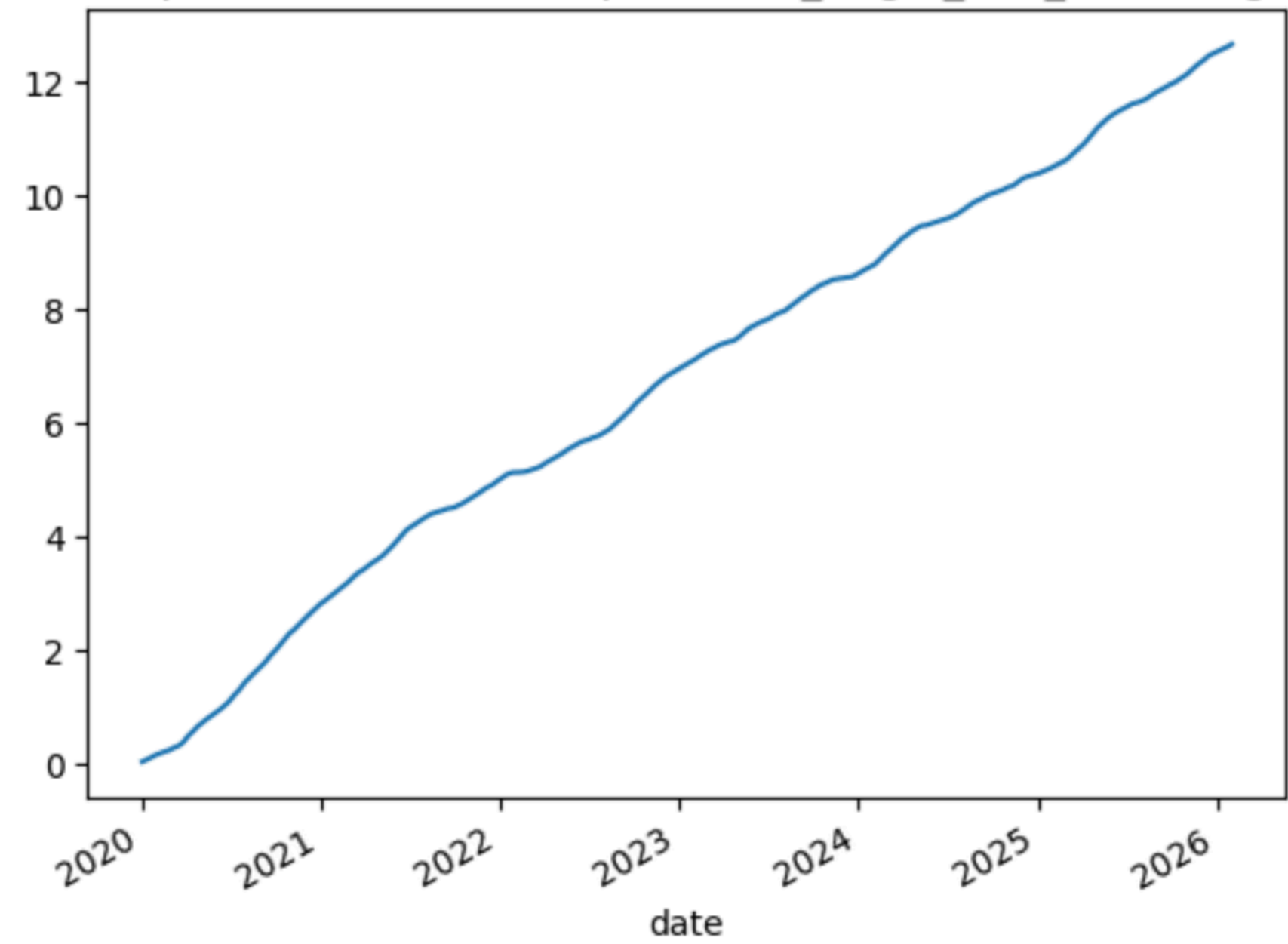
日本ユニバース以外の値の補完



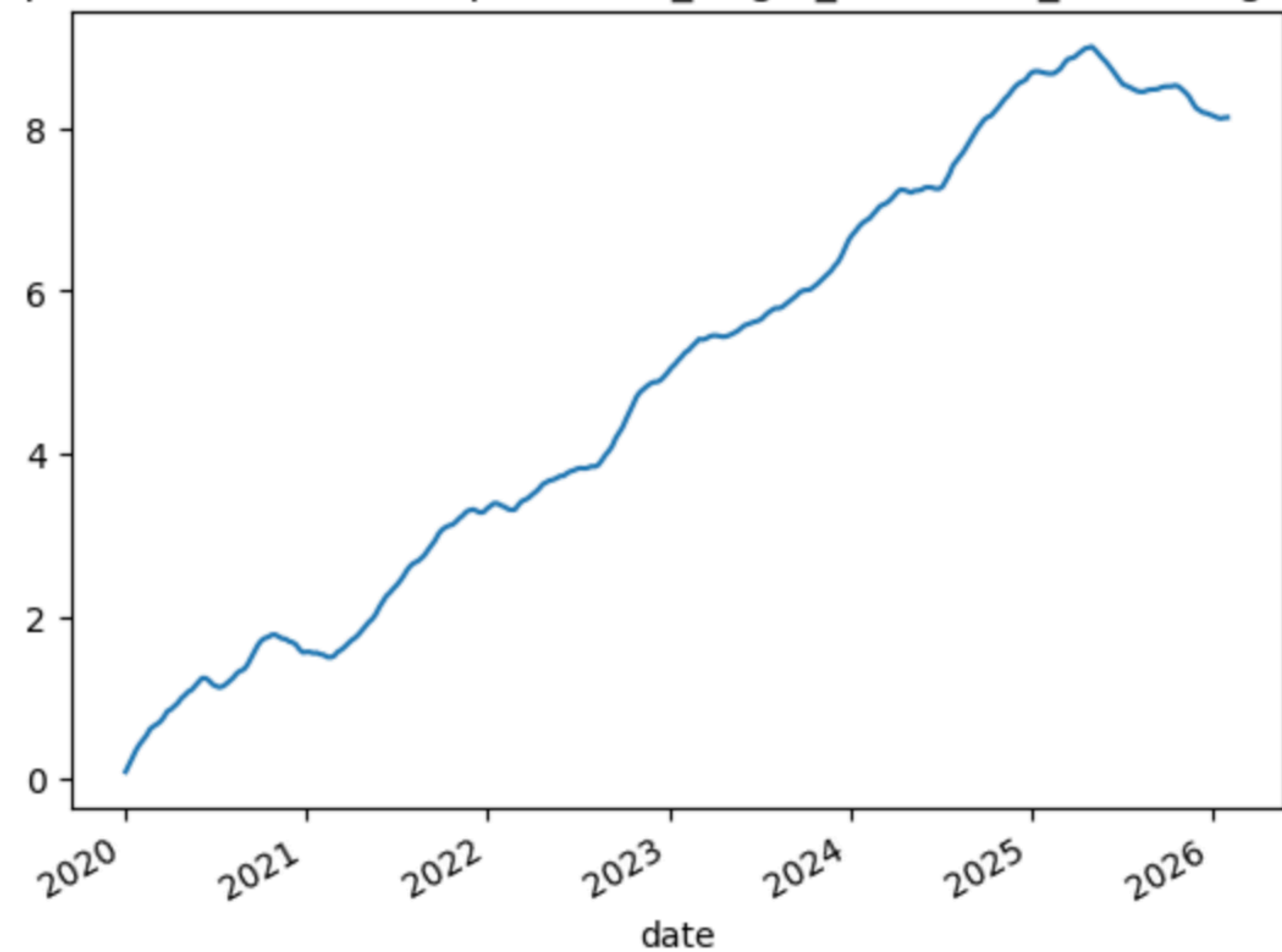
Current Model Development & Backtest Results

Target-Specific Performance: Successfully predicting target_chili_used for Alpha computation.

Cumulative Spearman Correlation: prediction_target_chili_60 vs target_chili_60



Cumulative Spearman Correlation: prediction_target_alexandra_60 vs target_alexandra_60



Current Model Development & Backtest Results

- Integrating unique Japanese equity data led to a substantial improvement in the Alpha score (achieving a Sharpe Ratio of 2.06).
- Evaluated Model Variants:
 - A blend of target_chili_60 predictions and other target predictions.
 - target_chili_60 predictions only.
- Insight: Certain targets can yield high Alpha scores even if their direct correlation with predictions appears low.

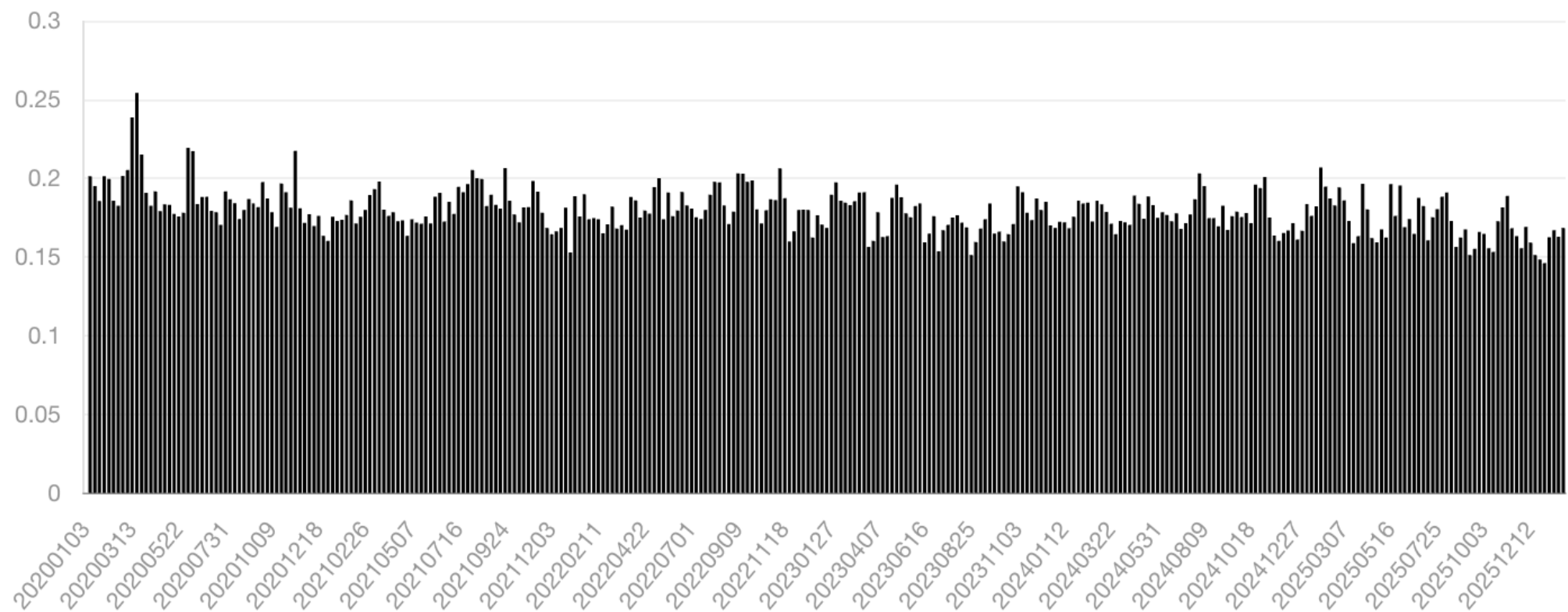
NUMERAI SIGNALS

SAKURA_ALPHA

Score	Mean	Standard Deviation	Sharpe	Max Drawdown
Alpha	0.0305	0.0148	2.0619	-0.0250
FNCv4	0.0073	0.0193	0.3804	-0.1441
RIC	0.0070	0.0289	0.2419	-0.5718
CORRv4	0.0051	0.0128	0.3957	-0.0907
ICv2	-0.0044	0.0760	-0.0575	-2.0225
Turnover	0.1795			

Performance

Turnover ▾

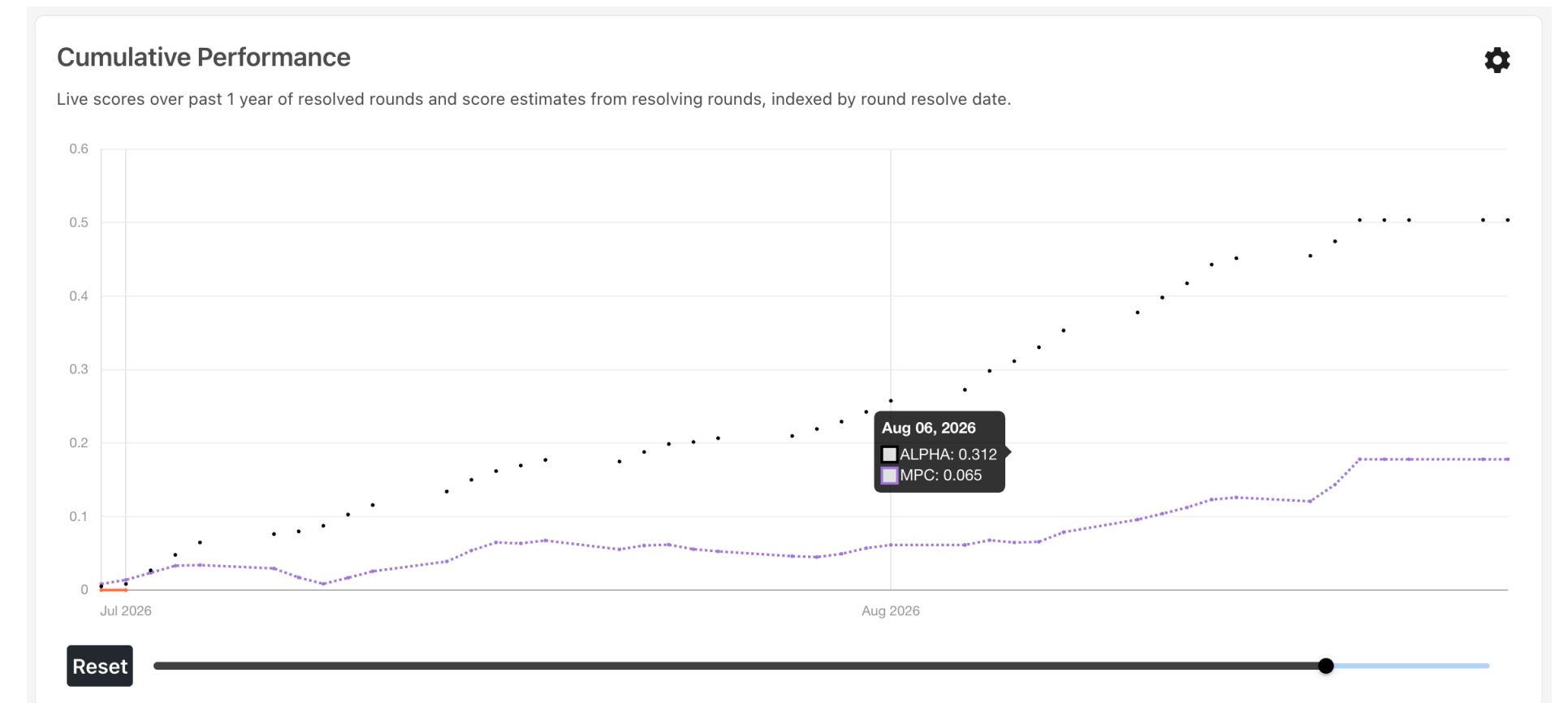


Current Model Results

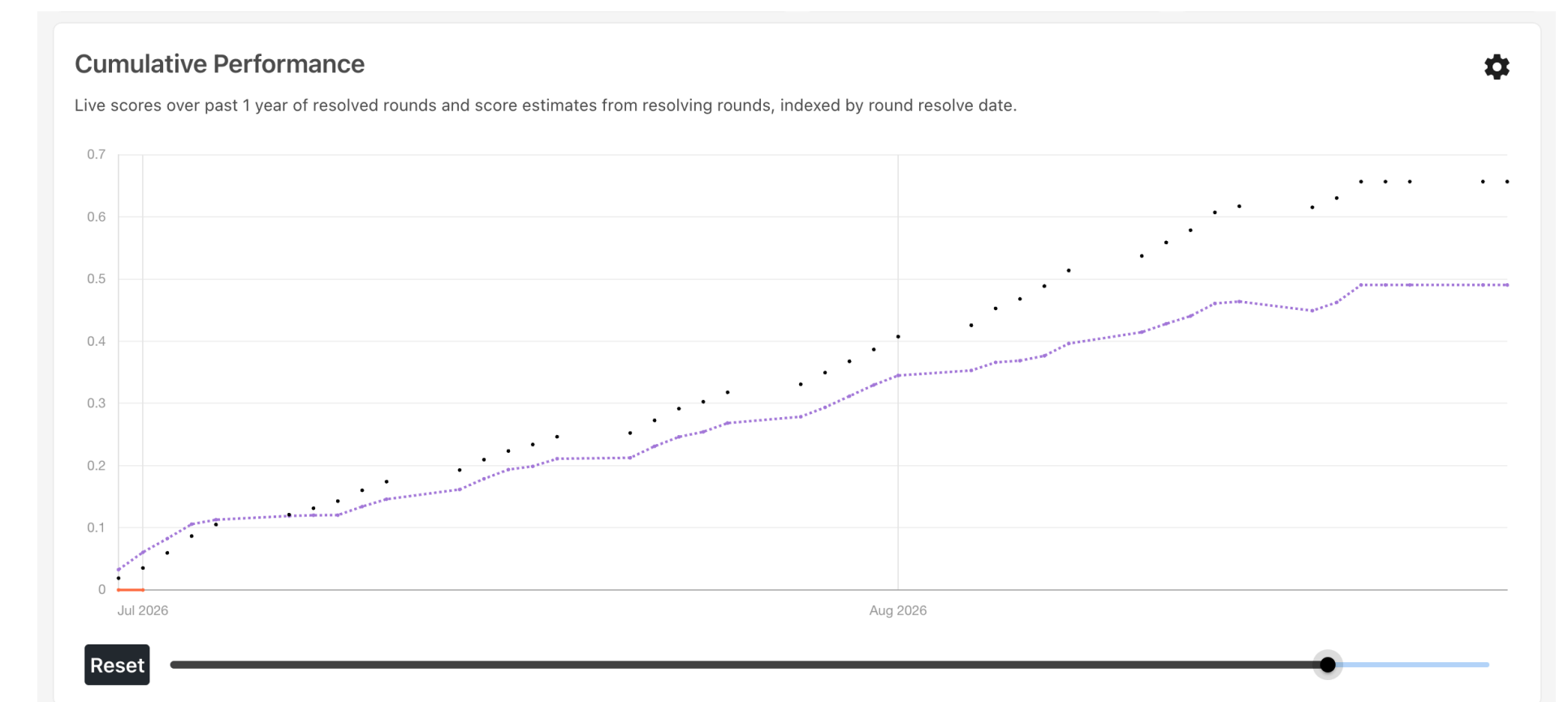
Performance Post-Implementation

- **target_chili + Other Targets:**
Showed clear signs of performance improvement.
- **target_chili only**
 - Performance improved dramatically. The target_chili_60-only variant—which maintained high correlation—emerged as the optimal model (e.g., achieving an Alpha of 0.312 and MPC of 0.065 in recent rounds).
- **target_chili only is the best!**

- target_chili + other_targets



- target_chili only

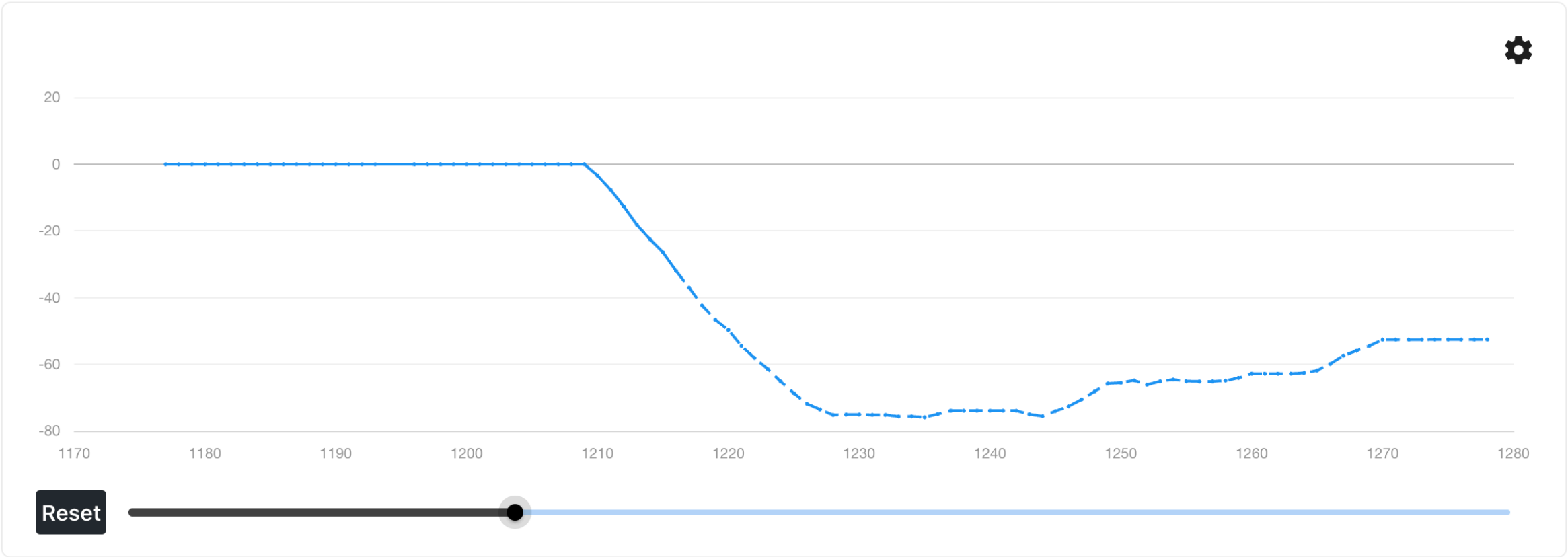


Current Payout Status & Summary

Live Payout Status (Current Drawdown due to Staking Concentration)

- **Staking Risk:** Concentrating stakes into a single model out of the three led to a significant temporary drawdown (-52.53 NMR).
- **Current Standings:** Currently, only FAIR_MARKET_ALPHA is in positive territory (+0.12 NMR).
- **Outlook:** The newly deployed model is performing strongly and is on track to break even and turn profitable. High expectations for upcoming rounds.

☐	Model	Total
☐	TOTAL	-52.53 NMR
☐	SAMURAI_ALPHA	-0.12 NMR
☐	FAIR_MARKET_ALPHA	+0.12 NMR
☐	SAKURA_ALPHA	-52.53 NMR
☐	LATE_TEST	0.00 NMR



Summary & Key Takeaways

- **The Beauty of Signals:** Numerai Signals offers far greater flexibility than the standard Tournament, turning proprietary alternative data into a true edge.
- **Beyond Simple Returns:** Generating a signal that merely correlates with raw returns is not enough; the core value lies in providing actionable predictions tailored to Numerai's portfolio constraints.
- **Key Architectural Wins:** Feature ranking coupled with a neutral 0.5 imputation for missing global data proved highly effective.
- **Optimal Configuration:** Currently, the target_chili-only variant yields the best performance. Integrating unique Japanese market insights has successfully unlocked hidden Alpha.
- **Next Steps: Wen USDC Staking?**
Planning to scale up position sizes (lot sizes) as soon as USDC staking is officially live!